Syllabus – ATSP 3135 – Interactive Art and Creative Coding – Fall 2026

Course Title: Interactive Art and Creative Coding
Semester: Fall 2026
Catalog Number: 3135 | **Class Number:** 1124 | **Section:** 001 | **Credits:** 3
Prerequisite: ATSP 2101 or permission of instructor
Location: MacLean 401
Time: Wednesdays, 9am-3pm (August 26-December 9)

Instructor: Douglas Rosman (he/him) (Preferred name: Doug)
Email: drosma@artic.edu (you can also send me a message in Canvas)
Office Location: MacLean 405
Office Hours: Available by appointment (send me an email to schedule)

Course Description

This studio course investigates the creative possibilities in programming, from interactivity to information visualization. Students explore interactive narratives and games, software art, simulations and emergent behaviors, and other code-based forms. Lectures and demonstrations provide a conceptual, aesthetic and technical foundation in programming as a creative practice. Techniques and concepts are presented through the open-source programming environment ~~Processing~~ p5.js, ~~with an introduction to advanced topics such as C++ and OpenFrameworks~~. Additionally, this course will explore web fundamentals (HTML and CSS), the version control tool git, as well as GitHub as a platform for open source code development.

Learning Goals

At the end of this course, students will...

- Attain an intermediate-level proficiency in creative coding as both a technical practice and conceptual framework, and be able to apply that knowledge to work critically with code as a medium for artistic expression and cultural intervention.
- Develop fluency in the languages and methodologies of creative coding in order to effectively articulate and contextualize computational approaches within contemporary art practice.

- Gain comfort utilizing both established and emerging open source tools to explore creative expression across digital media, including interactive installations, generative systems, web-based works, and experimental interfaces.
- Develop a critical understanding of the cultural and political dimensions of creative coding, including the communities that foster its development and the ways computational thinking shapes artistic discourse.
- Cultivate technical competency in core web technologies (HTML, CSS, JavaScript/p5.js) and development workflows (version control with GitHub) as foundations for experimental practice.

Student Learning Outcomes

At the end of this course, students will...

- **STUDIO SKILLS** Demonstrate proficient technical skills with technical control over materials, and/or a beginning use of more specialized tools and techniques in relation to creative coding and computational art.
- **MATERIAL EXPLORATION, FORM AND CONTENT** Research and apply technical approaches that are appropriate for their projects and/or research. Demonstrate a developing understanding of how computational processes relate to aesthetic outcomes; considers sustainability and environmental impact of digital practices.
- **HISTORICAL AND/OR THEORETICAL CONTEXT** Demonstrate a deepened knowledge of specialized histories and theories of creative coding, computational art, and digital culture in relationship to their practice.
- **CREATIVE PRACTICE RESEARCH** Practice the concepts and principles of research using open source tools, documentation, and community resources to create a framework for independent discovery, evaluation, and analysis; demonstrate an understanding of the questions, methods and objectives of creative coding research; communicate outcomes through code, documentation, or presentations.
- **INTERDISCIPLINARY CONTEXT** Apply concepts, theories, discourses, media, processes, technologies from distinct disciplines to the practice of creative coding to explore a research question or creative inquiry that bridges computation with other fields.
- **DIVERSE PERSPECTIVES** Participates in conversations and activities that reinforce the importance of diverse perspectives to enrich computational art and tech communities; demonstrates a developing self awareness of personal assumptions and biases about who belongs in creative coding spaces.
- **PRESENTATION** Purposefully present, project, perform, publish, install, mount or exhibit their computational work and articulate ideas in a manner that is clear and well organized; demonstrates an awareness of audience interaction and user experience; (when applicable)

adheres to the guidelines or requirements of digital platforms or installation spaces with limited oversight.
- **CRITIQUE** Engage in active listening and participate in discussion/reflection using specialized vocabulary from both technical and artistic domains; analyze and interpret feedback with sensitivity to different technical skill levels and creative approaches. Give informed and meaningful feedback to others about both conceptual and technical aspects of computational work and contribute positively overall to the critique process.

Receiving Credit for this Course

- Complete all required assignments, including a Final Project. All assignments that are required will be clearly listed as "REQUIRED" in Canvas
- Satisfy the SAIC attendance policy (miss no more than 2 class sessions)
- Contribute at least 5 resources to the class Are.na page

Other weekly assignments are strongly encouraged, but not required for course credit

Being on time and ready for class

- Please arrive at 9:00am to give yourself time to settle in
- Lectures will begin promptly at 9:05am. By 9:05, you should:
  - save and close any program, project or assignment unrelated to this class
  - have something ready for note taking
  - make sure your computer has enough battery, or that your charger is plugged in

Course Materials

- https://saic.instructure.com/courses/7489900 (class Canvas page. Assignments and important material will be posted here)

Readings

Readings will be uploaded to Canvas, and can be accessed from there.

- Code as Creative Medium: A Handbook for Computational Art and Design(2021), Golan Levin and Tega Brain
- The Nature of Code: Simulating Natural Systems with JavaScript (2024), Daniel Shiffman
- Getting Started with p5.js (2015), Lauren McCarthy, Casey Reas and Ben Fry

Course Schedule

The material in this class is subject to change based on the needs and impulses of the class. Assignment details will be published to Canvas when they are assigned.

Week 1 8/26 - Introductions + creative code review

Introductions, course overview, tool setup, intro to HTML and CSS, p5.js review
Assignment: create your home page
Read: *Getting Started with p5.js* 1/Hello, *Code as Creative Medium: A Handbook for Computational Art and Design* Introduction

Week 2 9/02 - p5.js bootcamp

Variables, program flow, conditional statements (if-statements), for-loops, functions, mouse input, randomness
Assignment: Iteration sketches

Week 3 9/09 - Objects, arrays and vectors, state machines

Creating, accessing and looping through arrays. Using vectors for position. Managing state with "state machines"
Assignment: Conditional testing and state machines

Week 4 9/16 - OOP (Object Oriented Programming) part 1: classes and object properties

Particle systems, objects, classes
Assignment: Midterm project proposal (create a living environment populated with entities that interact with each other)

Week 5 9/23 - OOP part 2: methods and behaviors, randomness and noise

Making objects change over time and interact with the world.
p5.sound library
Assignment: work on midterm

Week 6 9/30 - Midterm open studio time

Open Studio Time
Make sure midterm project is working with GitHub / GitHub Pages
Assignment: work on midterm

Week 7 10/07 - Midterm critiques

Midterm Critiques
Assignment: artist research presentation

Week 8 10/14 - Sound in p5.js, 3D in p5.js

Sound synthesis and playback, audio visualization, microphone input, 3D with WebGL in p5.js

Week 9 10/21 - Expanded interactivity: body tracking with ml5.js

Pose, face, body, hand tracking with ml5.js, present artist research project
Assignment: teach the class about a p5.js library

Week 10 10/28 - Drawing machines, p5.plotSvg and pen plotters

Code sketching for the pen plotter, working with the p5.plotSvg library, drawing with a pen plotter
Assignment: plot a drawing with the plotter in the io Lab

Week 11 11/04 - Working with Data / APIs, Final Project Assigned

Using publicly available data to drive your sketches, present "teach the class about a p5.js library", assign final project
Assignment: Final project proposal

Week 12 11/11 - Open Studio

Final project proposal presentations, open studio time

Week 13 11/18 - Open Studio Time

Open studio time to work on final projects, decide critique schedule, creative coding beyond p5.js

Week 14 11/25 - Thanksgiving Break (No Class)

Week 15 12/02 - Critique Week (No Class)

Week 16 12/09 - Final Project Critiques

SAIC Attendance and Participation Policy

Students are expected to attend all classes regularly and arrive on time. Full participation is required in all aspects of a course—including in-person sessions, synchronous online classes, and independent study.

Instructors may permit students up to two absences per semester—inclusive of absences due to illness and/or family/personal emergencies. If a student registers late during the add/drop period, they are responsible for catching up on all missed content and assignments; instructors may count those missed classes as absences.

Instructors have the discretion to define expectations around student tardiness, including policies related to late arrivals, mid-class breaks, and early departures. These expectations should be clearly outlined in the course syllabus and communicated to students during the first class to ensure transparency and consistency.

Instructors grant credit only to students who meet the standards and expectations of the course.

Students who are ill and unable to attend class should notify their instructor(s) by email or leave a message with the department office on the day they are absent. For extended absences due to illness, the student should contact Health Services, who will notify instructors. For other extenuating circumstances, students should reach out to the Academic Advising office. Please note that the written notification does not excuse a student from classes.

Federal Financial Aid Attendance Requirement: Recipients of federal aid must have begun attendance in classes for which their eligibility is based upon at the time of disbursement and, in the case of Federal Direct Loans (Subsidized, Unsubsidized, and PLUS), be enrolled at least half-time.

Religious Holiday Observance: SAIC recognizes the diverse religious and cultural practices of our community. Students are expected to notify their instructors early in the semester to discuss reasonable accommodations for holidays they observe.

SAIC Audit Policy

SAIC does not generally encourage degree seeking students to audit courses but does permit them to do so in some circumstances. Students are required to meet specific criteria and then additionally have the permission of the faculty to do so. Review the full Audit Policy here.

1. If you are teaching a **studio course**, students **are not permitted to audit.** Suggested language for your syllabus:

Audit Policy*: Per SAIC's Policy on non-credit enrollment, auditing this course is not permitted. Students wishing to participate in this course must be enrolled for credit.*

Class Progress Reports (CPRs)

Class Progress Reports (CPRs) are used to communicate between faculty, students, and academic advisors to give feedback on class performance. You may receive a CPR if you have missed class, or missed an assignment, critique, etc. These communications are intended to help you understand what you need to do to stay on track or get back on track and succeed in this class.

Accommodations for Students with Disabilities

SAIC is fully committed to complying with all laws ensuring equal opportunities for students with disabilities. Students with known or suspected disabilities are encouraged to contact the Disability and Learning Resource Center (DLRC) to schedule a virtual appointment. During this appointment, DLRC staff will review the student's documentation and work collaboratively to determine reasonable accommodations.

Once accommodations are approved, the DLRC will email a letter outlining the accommodations to both the student and their instructors. Students should be aware that not all approved accommodations will be applicable to every course. It is recommended that students speak with each instructor to discuss how their accommodations may be applied in the context of each class before they are implemented. To ensure timely support, students should contact the DLRC as early in the semester as possible. The DLRC can be reached via phone at 312.499.4278 or email at dlrc@saic.edu.

Student Responsibility in the use of Accommodations

Students approved for accommodations through the DLRC are responsible for communicating with their instructors regarding the use and application of those accommodations. The implementation of accommodations requires an interactive process between the student and the instructor to determine how accommodations might be applied in the context of each specific course. Because course content, assignments, and learning objectives vary, accommodations must be considered on a course-by-course basis to ensure that they are both appropriate and do not compromise essential learning outcomes, which students are still expected to meet.

Accommodations do not excuse students from the standard requirements of communication and attendance. If a student needs to miss class or modify participation due to an approved accommodation, they must communicate with the instructor in advance whenever possible. In situations where advance notice is not feasible, students must notify the instructor within 24 hours of the missed class or academic activity. Failure on the student's part to communicate in a timely manner may result in the inability to apply the accommodation for that specific instance.

Writing Center

Tutors are available in person and online to help students achieve their writing goals at any stage of their writing process. All students are welcome, and they can work on essays, artist statements, application materials, presentation texts, theses, proposals, creative writing, or social media posts. The Writing Center tutors are kind, encouraging, and interested!

Writing Center Hours (CST):
Monday – Thursday: 9:15 AM - 7:15 PM and
Friday: 9:15 AM - 5:15 PM

Though drop-ins are welcome, the best way to guarantee an appointment is to schedule one via Navigate: https://www.saic.edu/academics/writing-center

116 S. Michigan Ave., 10th Floor (Lakeview building)
Phone: 312-499-4138

Academic Misconduct

From the [SAIC Student Handbook](), page 12

Academic misconduct includes both plagiarism and cheating, and may consist of: the submission of the work of another as one's own; unauthorized assistance (as defined by individual instructors and laid out in the course syllabus) on a test or assignment; submission of the same work for more than one class without the knowledge and consent of all instructors; or the failure to properly cite texts or ideas from other sources. Academic misconduct also includes the falsification of academic or student-related records, such as transcripts, evaluations, and letters of recommendation.

Academic misconduct extends to all spaces on campus, including satellite locations and online education.

Academic integrity is expected in all coursework, including online learning. It is assumed that the person receiving the credit for the course is the person completing the work. SAIC has processes in place, including LDAP authentication, to verify student identity.

Classroom Policy on AI-generated Code

Use of Generative AI tools specifically to write code is prohibited unless stated otherwise on a per-assignment basis. You are allowed to use AI tools as a research or learning aid, but not to generate your code.

Generative AI tools, particularly Large Language Models (LLMs) like ChatGPT, Claude or Gemini can generate functioning code in a variety of languages, simply by asking them for what you want in human language. Professional coders at all levels of industry are building AI into their workflows to increase their productivity. In fact, some are forecasting that coders are some of the workers most vulnerable to [having their jobs replaced by AI in the near future](). *Even I use AI to help me write code for my own creative projects.*

So, why am not allowing you to use them for your coding assignments? I have practical and philosophical reasons for this:

1. Relying on AI to generate code prevents you from actually learning how to code.
2. Understanding coding fundamentals is necessary for you to use AI code assistance effectively (if an AI generates broken code, you probably won't know how to fix it!)
3. Code itself is an expressive medium, and has value outside of simply what it *does*.
4. Learning to code teaches you how your computer works.

Support Resources for Students

The Office of Students Affairs is here to help students achieve success in and outside of the classroom and studios. Staff members are available to assist students with a wide-range of issues and concerns, including mental and physical health concerns, food and housing insecurity, conflicts with others, and much more. We are available during typical business hours (9-5pm, Mon-Friday); however, we also have staff available after-hours to address emergency concerns.

In case of an emergency, please contact SAIC Campus Security, 24 hours a day, by visiting any campus security desk or calling 312.899.1230. They can assist you and/or connect you with a staff member who can provide support for you.

Food and Housing Resources

If you have difficulty affording groceries or accessing food every day, and/or do not have a safe and stable place to live, please contact the Office of Student Affairs - (312) 629-6800 / studenthelp@saic.edu during business hours. If you contact them after hours, someone will respond the next business day. You can also find links and resources at this site, curated by Student Affairs: Student Support Resources and Information.

SAIC Food Pantry

Spoonful Food Pantry is available to current SAIC students who are experiencing difficulty accessing food because of a financial emergency or ongoing constraints. Students can request a pre-packaged bag of non-perishable groceries (vegan and gluten free available) by completing the online form. Once approved, students will receive a pre-packaged bag of non-perishable groceries.

Wellness Center

The SAIC Wellness Center, which includes Counseling Services, Health Services, and the Disability and Learning Resource Center, is also here to support students' mental health, physical health, and accessibility needs. You may contact them at:

Counseling Services: counselingservices@saic.edu and 312-499-4271 (press 1 to speak to a counselor after hours)

Health Services: healthservices@saic.edu and 312-499-4288 (After hours contact the 24-Hour Nurse Line at 877-924-7758)

Disability and Learning Resource Center: dlrc@saic.edu and 312-499-4278

Ombudsperson

The services of the SAIC Ombudsperson are available to all students, faculty, and staff at SAIC. The Ombuds is a confidential, neutral, independent and informal resource; you may speak with them about any conflict you are encountering at SAIC. The Ombuds will help you explore the conflict and consider options for resolution. To learn more or to contact the Ombuds, visit saic.edu/ombuds.

Statement on Academic Freedom

The School of the Art Institute of Chicago is a community of educators, students, and staff whose artistic, design, and scholarly work is characterized, among others, by an ethos of intellectual and imaginative curiosity; the love and production of knowledge, art, and design; and the joy of creating. This ethos can be sustained, and the above learning goals can be achieved, only in an institutional and cultural framework of academic freedom, freedom of expression, and equality; these are the conditions within which learning, research, and creative output flourish. Such a framework also allows members of communities whose speech has historically been silenced to fully and equally participate in the same free expression that has historically been the privilege of only some segments of society. This framework also helps us navigate through conflict and tension—themselves vital aspects of educational, creative, and intellectual growth—and it helps us differentiate between the concepts of tension or offense on the one hand, and those of harm, discrimination, and harassment on the other.

For the full version of the statement, please visit the Faculty Resources webpage under Faculty Handbook and Guides > All Faculty. The statement can be found in Section 9 - STATEMENT ON ACADEMIC FREEDOM AND FREEDOM OF EXPRESSION (page 17).